

JKBOSE 2026

CLASS 11TH



Cell Cycle And Cell Division

LECTURE-1

ZOOLOGY

DAVOOD SIR (CEO of NG)

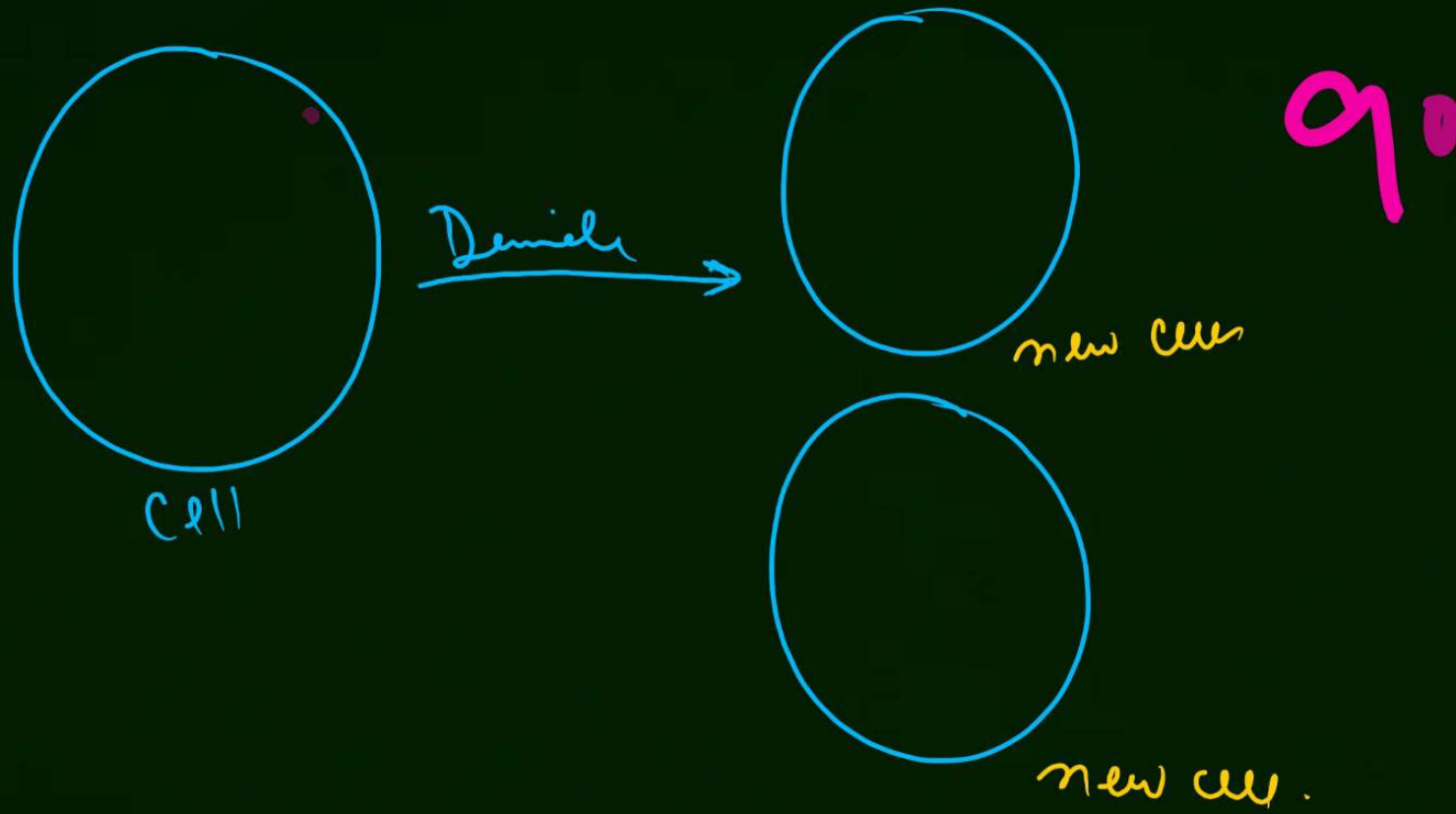


Physics Wallah

Topics To Be Covered

- 1** Introduction to Cell Cycle
- 2** Phases of cell cycle
- 3** Significance of Mitosis
- 4** Previous Year NEET MCQ'S

Cell Cycle



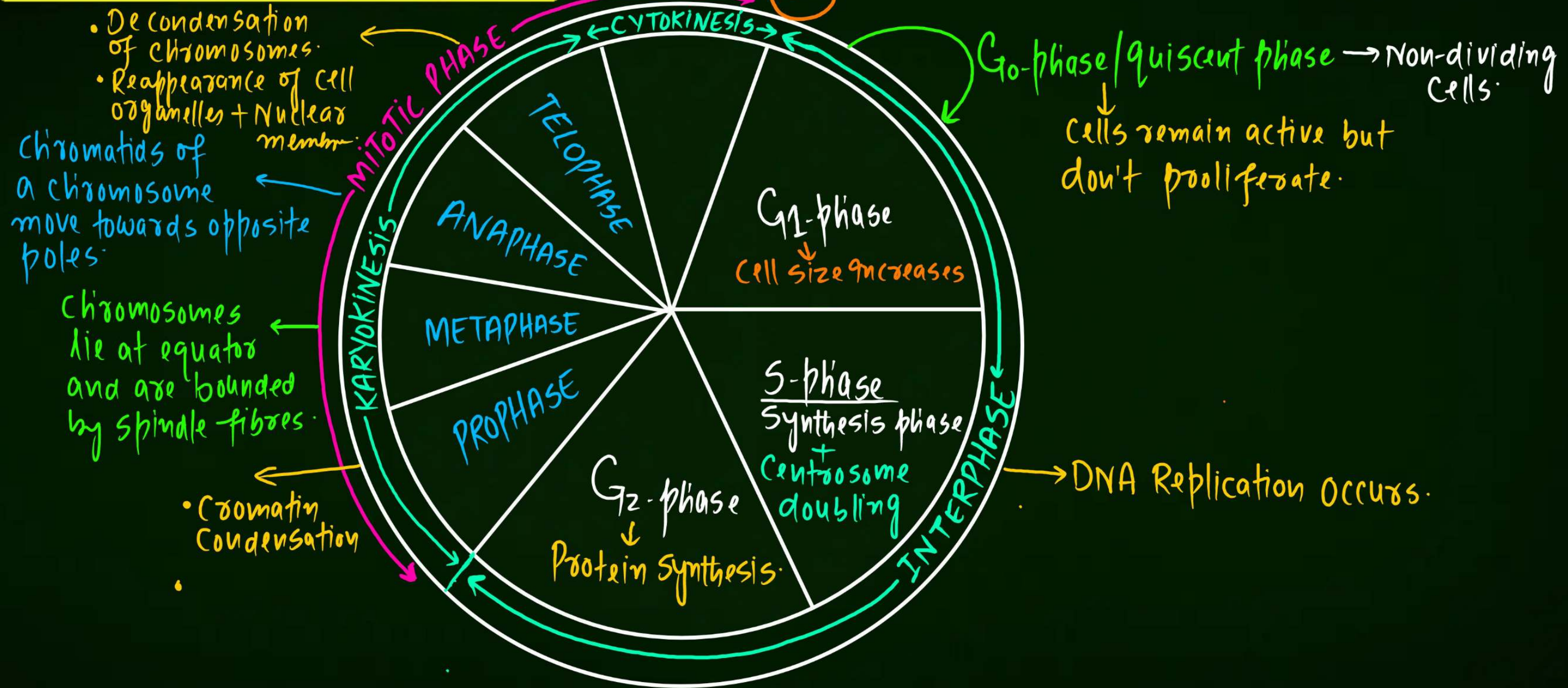
Cell Cycle

The cell cycle is the sequence of events by which a cell duplicates its genome, synthesizes other cellular components, and divides into two daughter cells.

While cell growth (cytoplasmic increase) is continuous, DNA synthesis happens only during a specific stage of the cell cycle. The replicated DNA (chromosomes) is then distributed to daughter nuclei through a complex, genetically controlled process during cell division.

- A typical human cells divide once in approximately every 24 hours.
- Yeast can progress through the cell cycle in only about 90 minutes.

Phases of Cell Cycle



Phases of Cell Cycle

The cell cycle is divided into two basic phases:

① Interphase

- Takes 23 hours.
- Also called as Resting phase as no such change is visible.
- Interphase is divided into three sub stages.
 - ① G₁ phase
 - ② S-phase
 - ③ G₂ phase.

② M Phase (Mitosis phase)

Interphase of Cell Cycle

The M Phase is when actual cell division (mitosis) occurs, while interphase is the phase between two M Phases. In the 24-hour human cell cycle, cell division lasts about 1 hour, and interphase takes up over 95% of the cycle. The M Phase begins with karyokinesis (nuclear division) and usually ends with cytokinesis (cytoplasmic division). Though called a resting phase, interphase involves cell growth and DNA replication in preparation for division. It is subdivided into three phases:

G1 phase (Gap 1)

S phase (Synthesis)

G2 phase (Gap 2)

Interphase of Cell Cycle

G1 Phase:- Interval between mitosis and start of DNA replication. The Cell is metabolically active and grows continuously but does not replicate DNA.

Some adult cells (e.g., heart cells) do not divide others divide occasionally to replace lost/damaged cells. These heart cells exit G1 into a resting, non-dividing stage called the quiescent stage (G0). Cells in G0 remain metabolically active but do not proliferate unless needed.

S Phase (Synthesis):- DNA synthesis/replication occurs, DNA amount doubles but not chromosome. In animal cells, DNA replicates, and centriole duplicates in the cytoplasm.

G2 Phase:- Proteins are synthesized, cell growth continues in preparation for mitosis.

In animals, mitotic cell division typically occurs only in diploid somatic cells. An exception is seen in male honey bees, where haploid cells undergo mitosis. Plants can perform mitotic divisions in both haploid and diploid cells.

Question [NEET-2023]

Kaun Banayga Nabkaar



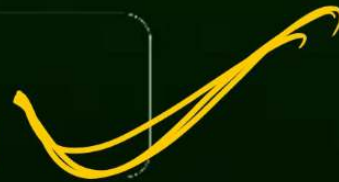
Among eukaryotes, replication of DNA takes place

1 G_1 phase

2 G_2 phase

3 M phase

4 S phase.



Question [NEET-2023]

Kaun Banayga Nabkaar



Match List I with List II.

List-I

A. M Phase

B. G_2 Phase

C. Quiescent stage

D. G_1 Phase

List II

I. Proteins are synthesized

II. Inactive phase

III. Interval between mitosis and initiation of DNA replication

IV. Equational division

Choose the correct answer from the options given below.

1 A-IV, B-I, C-II, D-III

2 A-III, B-II, C-IV, D-I

3 A-II, B-IV, C-I, D-III

4 A-IV, B-II, C-I, D-III

Question [NEET-2023]

Kaun Banayga Nabkaar



Which one of the following is the quiescent stage of cell cycle?

1 M

2 G_2

3 G_1

4 G_0

M-Phases of Cell Cycle

Mitosis is the **most dramatic phase** of the cell cycle, involving major reorganisation of nearly all cellular components. Since the chromosome number remains same between parent and progeny cells, it is called an **equational division**. Mitosis has been divided into four stages specifically karyokinesis. The four stages of karyokinesis are:

Prophase

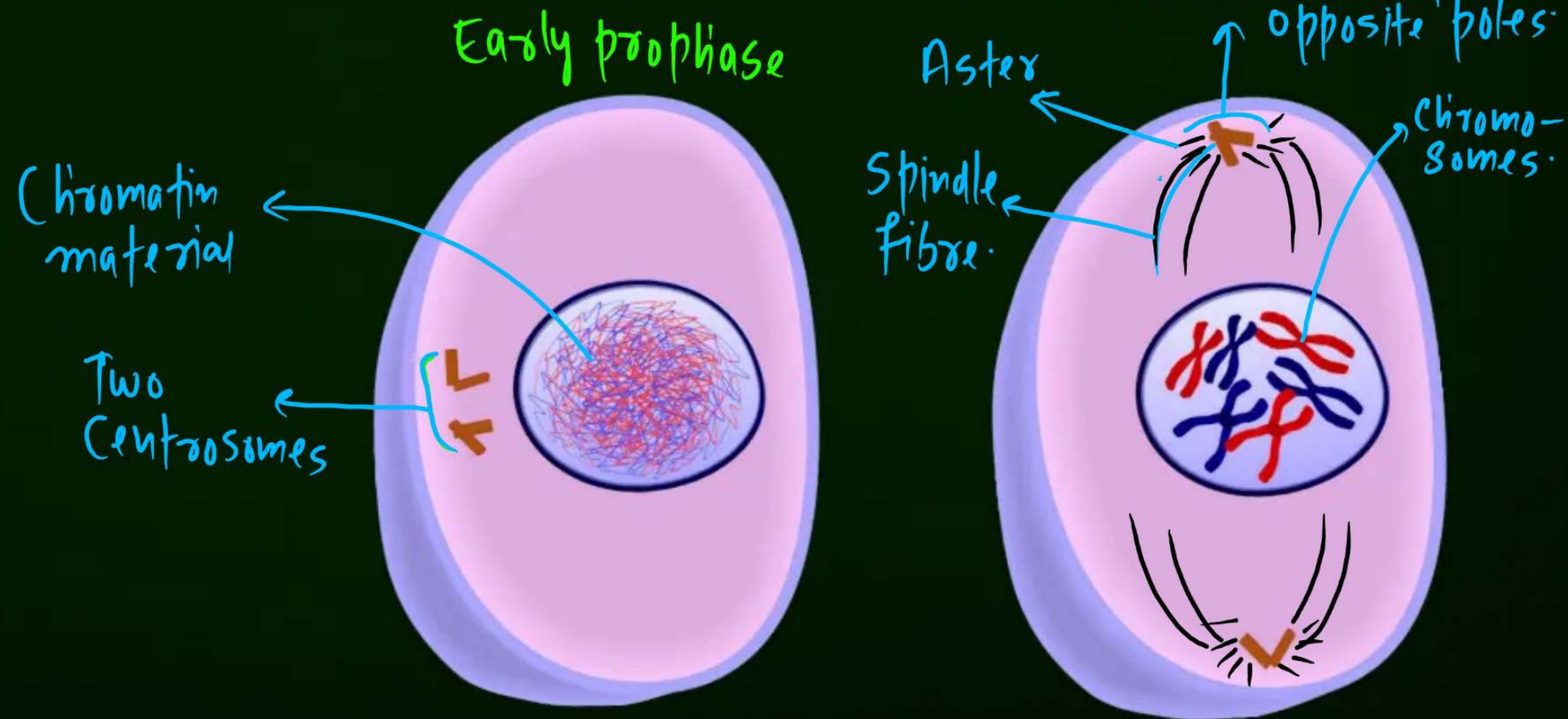
Metaphase

Anaphase

Telophase

Prophase

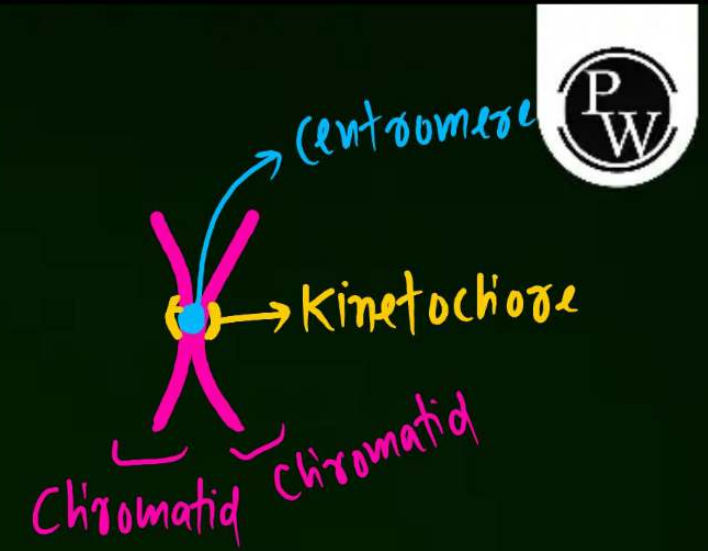
- Chromatin material Condenses into chromosomes.
- Two Centrosomes move to opposite poles.
- Centrosomes radiate asters + Spindle fibres (Centrosome + Aster + Spindle fibre → Spindle Apparatus)
- Cell organelles disappear.



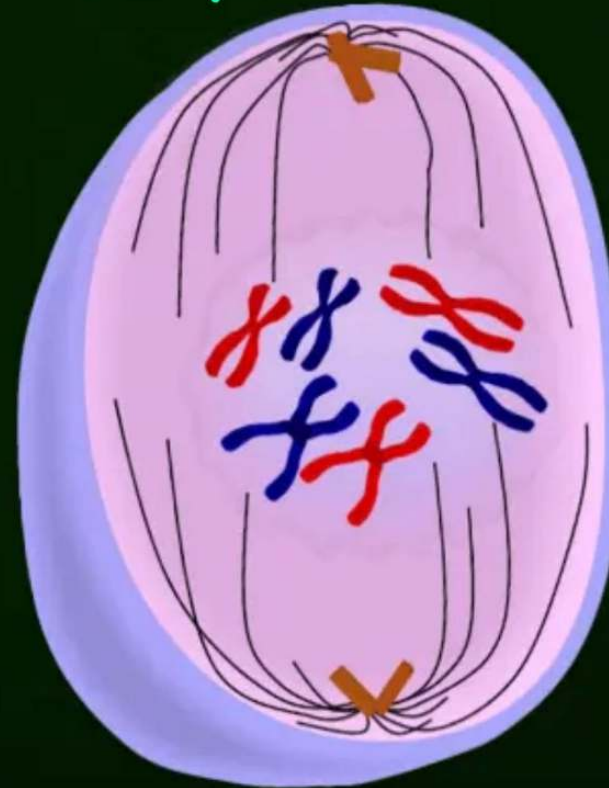
Prophase

- a) Prophase begins with the condensation of chromosomal material and becomes untangled. The centrosome, duplicated during the S phase, starts moving to opposite poles of the cell. Completion of prophase is marked by:
- I. Chromatin condenses into compact mitotic chromosomes, each with two chromatids joined at the centromere.
 - II. Each centrosome radiates microtubules called **asters**. Together with spindle fibres, they form the **mitotic apparatus**.
 - III. Golgi complexes, endoplasmic reticulum, nucleolus, and nuclear envelope disappear.

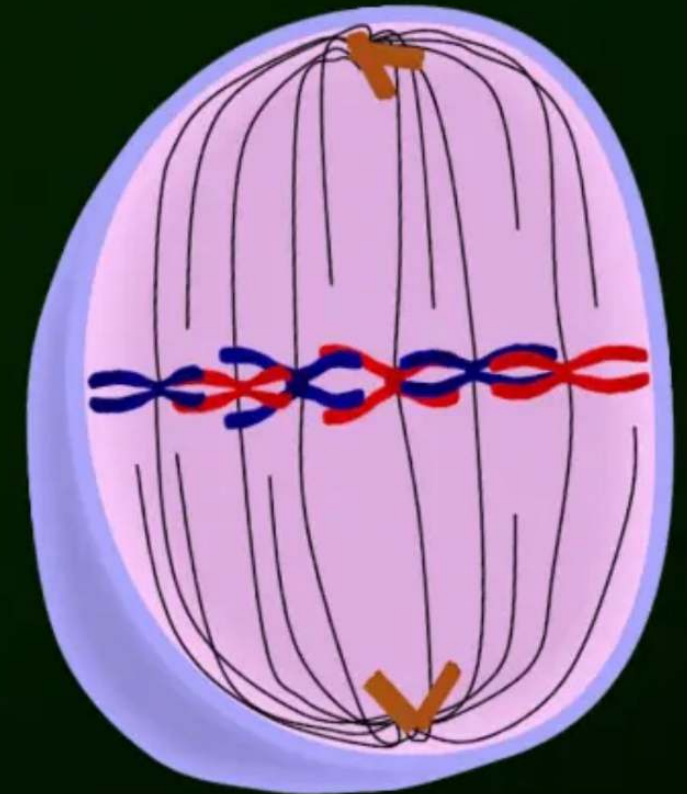
Metaphase



Early Metaphase



Late metaphase.



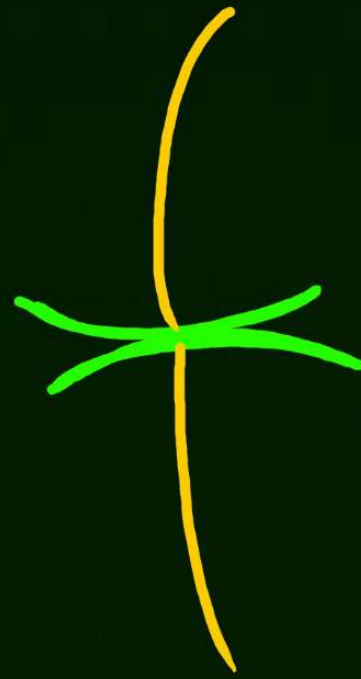
Metaphase

- In metaphase, the nuclear envelope fully disintegrates and the Chromosome condensation is completed, and each metaphase chromosome consists of two sister chromatids and at this stage morphology of chromosomes is mostly studied.
- At the centromeres, small disc-shaped structures called kinetochores are present.

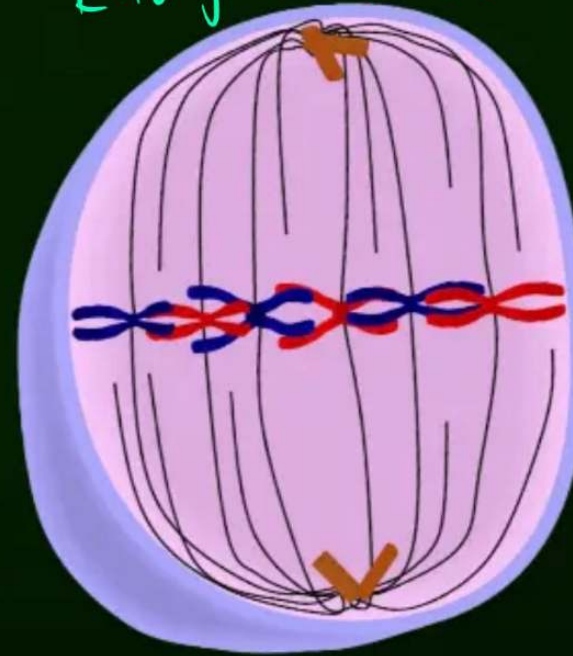
Metaphase is marked by:

- All chromosomes aligning at the equator (**metaphase plate**).
- One chromatid of each chromosome is connected by its kinetochore to spindle fibres from one pole, and the sister chromatid to spindle fibres from the opposite pole

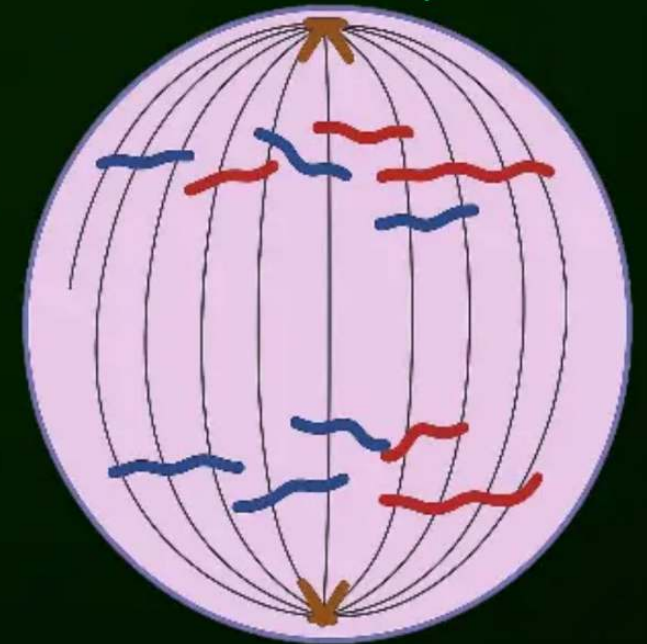
Anaphase



Early Anaphase



Late Anaphase



Anaphase

In anaphase, each chromosome at the metaphase plate splits simultaneously, and the resulting daughter chromatids now called daughter chromosomes of the future daughter nuclei or cell begin to migrate toward opposite poles of the cell and it is characterised by:

- Centromeres split and chromatids separate.
- Chromatids move to opposite poles.

Question [NEET-2025]

Kaun Banayga Nabkaar

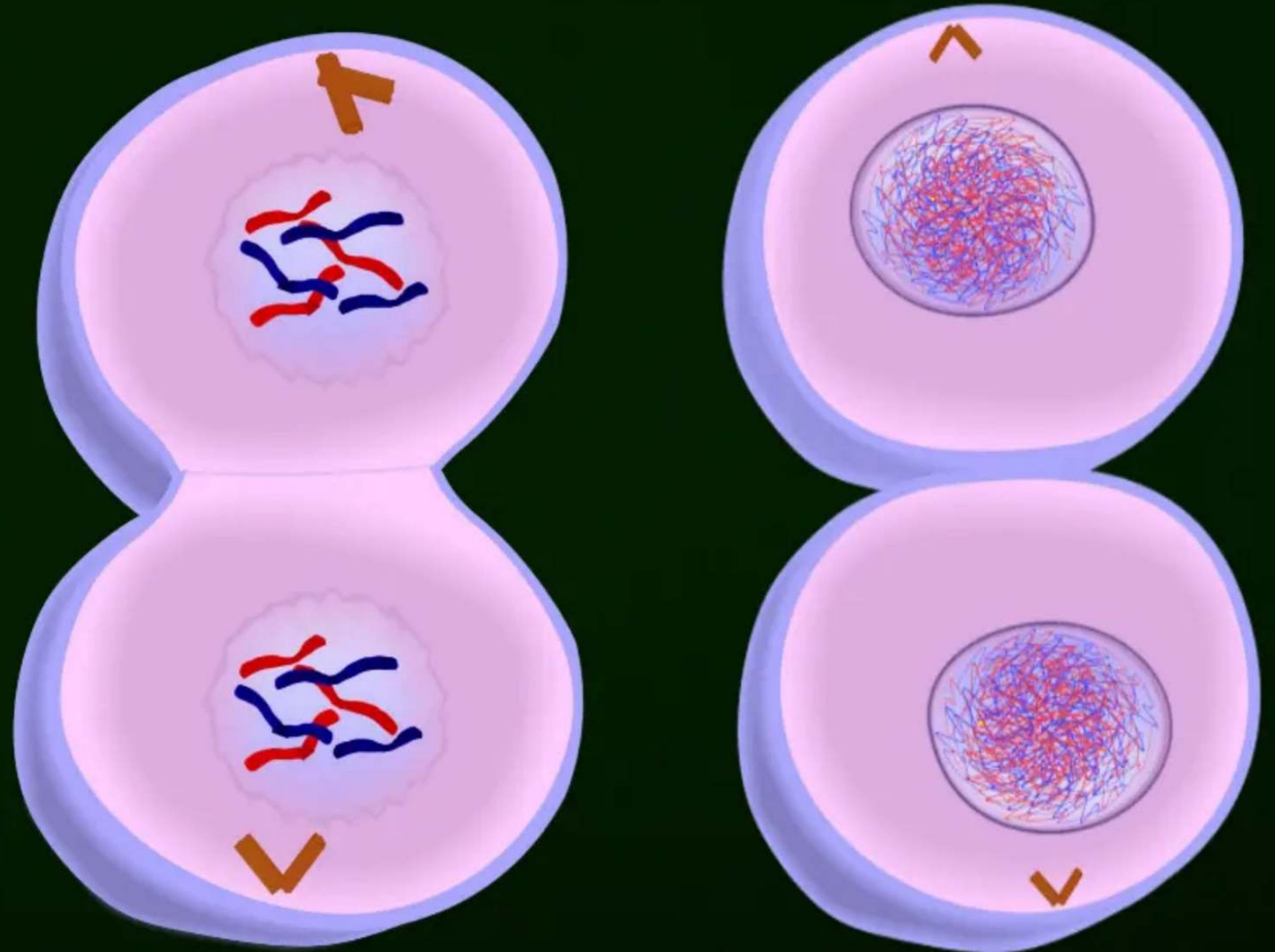


What is the main function of the spindle fibers during mitosis?

- 1** To synthesize new DNA
- 2** To repair damaged DNA
- 3** To regulate cell growth
- 4** To separate the chromosomes.

Telophase

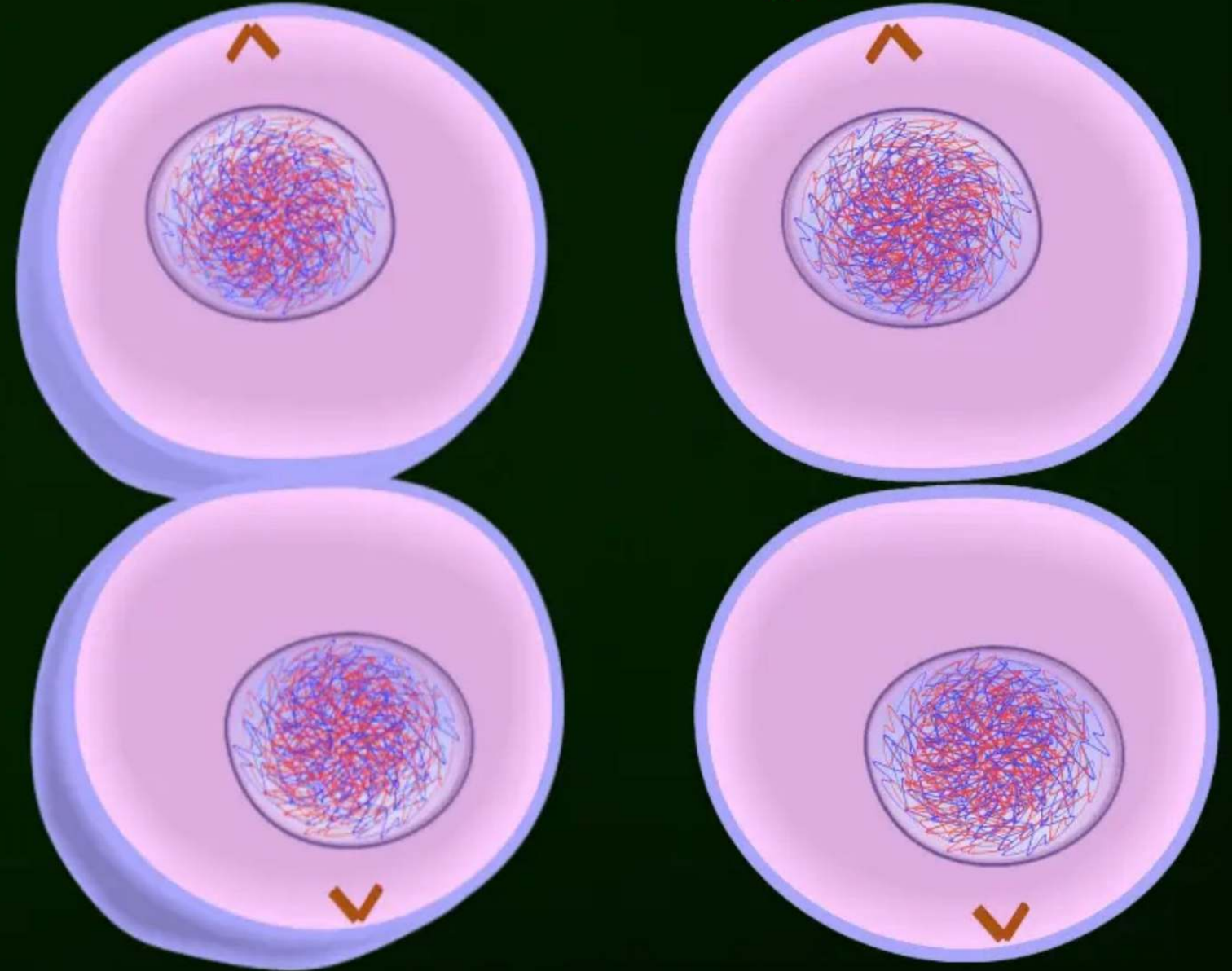
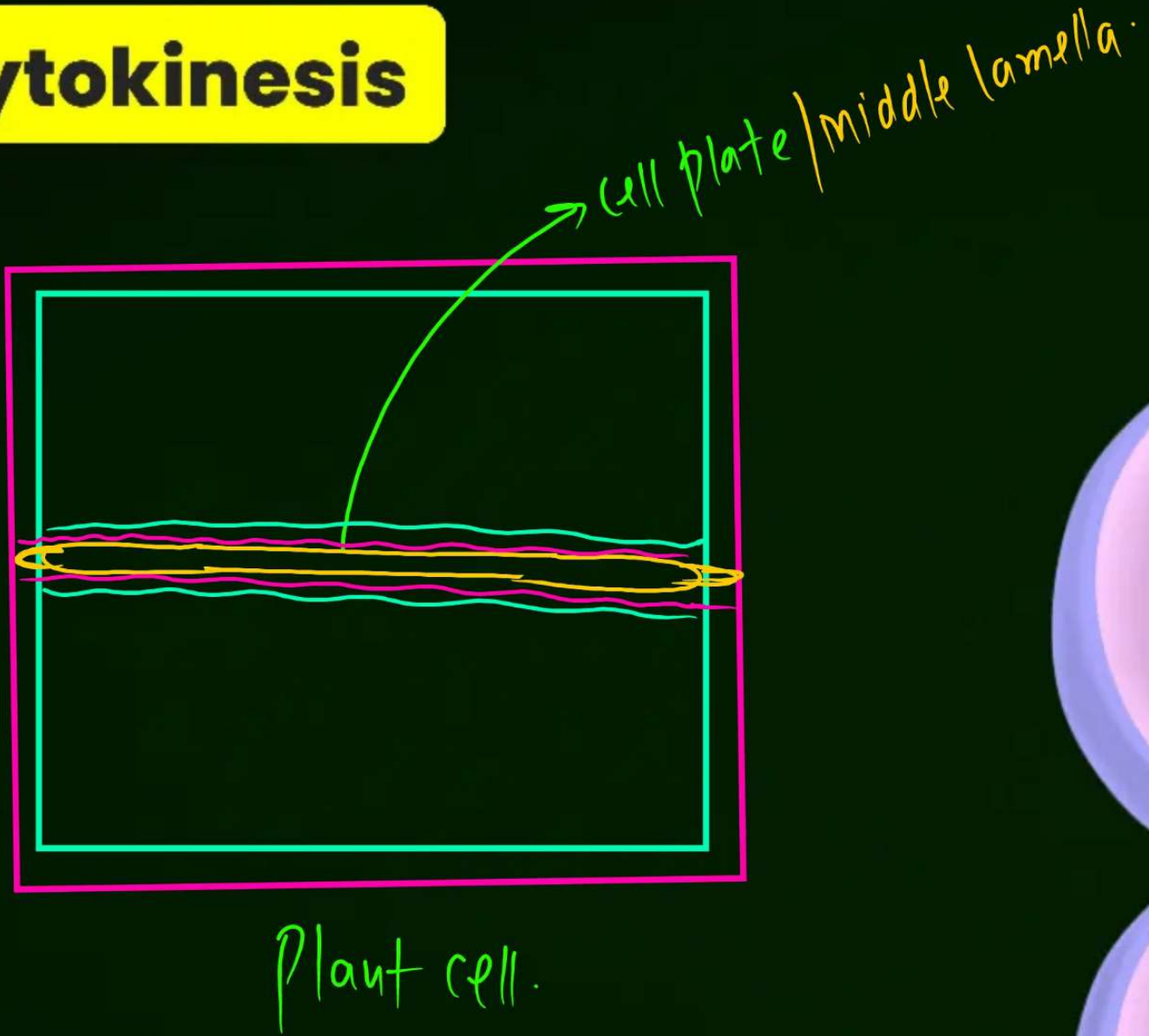
- Decondensation of Chromosomes into Chromatin.
- Nuclear membrane Reappears.
- Cell organelles reappears.



Telophase

- In telophase, **chromosomes at the poles begin to decondense**. A nuclear envelope forms around each chromosome cluster, **creating two daughter nuclei**. The nucleolus, **Golgi complex**, and **endoplasmic reticulum (ER)** are reformed

Cytokinesis



Cytokinesis

- Cytokinesis is the division of the cytoplasm, completing cell division and producing two daughter cells.
- In animal cells, it occurs through the formation of a cleavage furrow in the plasma membrane, which deepens and joins at the center, splitting the cytoplasm.
- In plant cells, cell wall formation begins in the center and extends outward. It starts with the formation of a **cell plate**, which becomes the middle lamella between the walls of the two new cells.
- In some organisms karyokinesis is not followed by cytokinesis as a result of which multinucleate condition arises leading to the formation of **syncytium** e.g: Liquid endosperm in coconut.

Question [NEET-2024]

Kaun Banayga Nabkaar

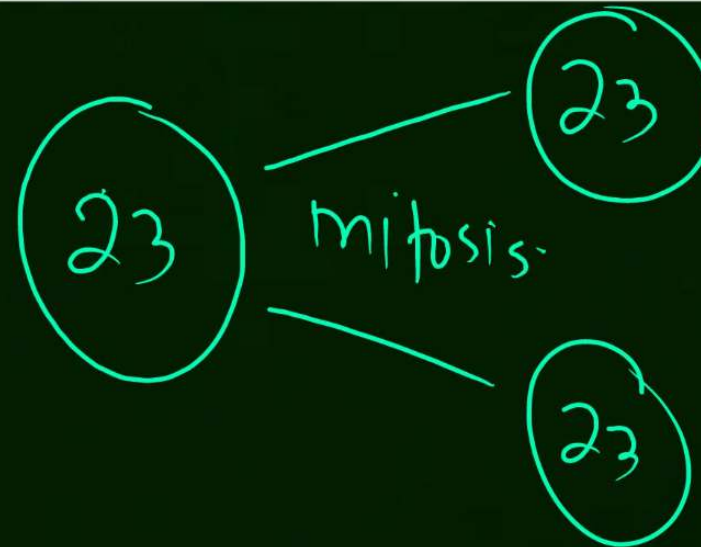


Following are the stages of cell division:

- ✓ A. Gap 2 phase
- ✓ B. Cytokinesis
- ✓ C. Synthesis phase
- ✓ D. Karyokinesis
- ✓ E. Gap 1 phase

Choose the correct sequence of stages from the options given below.

- ✗ **1** C-E-D-A-B
- ✗ **2** E-B-D-A-C
- ✗ **3** B-D-E-A-C
- 4** E-C-A-D-B. ✓



Significance of Mitosis

Mitosis (equational division) is typically limited to diploid cells, though haploid cells in some lower plants and social insects can also divide by mitosis. Mitosis is crucial for:

- a) Growth of multicellular organisms, restoring the nucleo-cytoplasmic ratio when cell growth disrupts it.
- b) Cell repair, as it replaces cells in the epidermis, gut lining, and blood.
- c) Continuous plant growth via mitosis in meristematic tissues (apical and lateral cambium), enabling plants to grow throughout their life.

Question [NEET-2024]

Kaun Banayga Nabkaar



Spindle fibres attach to kinetochores of chromosomes during:

1 Prophase

2 Metaphase.

3 Anaphase

4 Telophase

Thank

You